MultiLabelSoftMarginLoss#

https://pytorch.org/docs/stable/generated/torch.nn.MultiLabelSoftMarginLoss.

Description:

Creates a criterion that optimizes a multi-label one-versus-all loss based on max-entropy, between input xxx and target yyy of size (N,C)(N, C)(N,C). For each sample in the minibatch: where i∈{0, ⋯, x.nElement()−1}i \in \left\{0, \; \cdots , \; \text{x.nElement}() - 1\right\}i∈{0,⋯,x.nElement()−1}, y[i]∈{0, 1}y[i] \in \left\{0, \; 1\right\}y[i]∈{0,1}. weight (Tensor, optional) – a manual rescaling weight given to each class. If given, it has to be a Tensor of size C. Otherwise, it is treated as if having all ones. size_average (bool, optional) – Deprecated (see reduction). By default, the losses are averaged over each loss element in the batch. Note that for some losses, there are multiple elements per sample. If the field size_average is set to False, the losses are instead summed for each minibatch. Ignored when reduce is False. Default: True reduce (bool, optional) – Deprecated (see reduction). By default, the losses are averaged or summed over observations for each minibatch depending on size_average. When reduce is False, returns a loss per batch element instead and ignores size_average. Default: True

Function Signature

```
class torch.nn.MultiLabelSoftMarginLoss(weight=None,
size_average=None, reduce=None, reduction='mean')[source]#

Parameters

Shape:

forward(input, target)[source]#

Return type
```

Returns:

Tensor

Detailed Documentation

Documentation

Creates a criterion that optimizes a multi-label one-versus-all loss based on max-entropy, between input xxx and target yyy of size (N,C)(N, C)(N,C). For each sample in the minibatch:

where i∈{0, ⋯, x.nElement()−1}i \in \left\{0, \; \cdots , \; \text{x.nElement}() - 1\right\}i∈{0,⋯,x.nElement()−1}, y[i]∈{0, 1}y[i] \in \left\{0, \; 1\right\}y[i]∈{0,1}.

weight (Tensor, optional) – a manual rescaling weight given to each class. If given, it has to be a Tensor of size C. Otherwise, it is treated as if having all ones.

size_average (bool, optional) – Deprecated (see reduction). By default, the losses are averaged over each loss element in the batch. Note that for some losses, there are multiple elements per sample. If the field size_average is set to False, the losses are instead summed for each minibatch. Ignored when reduce is False. Default: True

reduce (bool, optional) – Deprecated (see reduction). By default, the losses are averaged or summed over observations for each minibatch depending on size_average. When reduce is False, returns a loss per batch element instead and ignores size_average. Default: True

reduction (str, optional) – Specifies the reduction to apply to the output: 'none' | 'mean' | 'sum'. 'none': no reduction will be applied, 'mean': the sum of the output will be divided by the number of elements in the output, 'sum': the output will be summed. Note: size_average and reduce are in the process of being deprecated, and in the meantime, specifying either of those two args will override reduction. Default: 'mean'

Input: (N,C)(N, C)(N,C) where N is the batch size and C is the number of classes.

Target: (N,C)(N, C)(N,C), label targets must have the same shape as the input.

Output: scalar. If reduction is 'none', then (N)(N)(N).

Runs the forward pass.
